Coloured Surfaces, Rooted Transitions, and Kempe Surgery for D_5-Flows: Structural Lemmas and Finite Evidence

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Abstract

The standard five-cycle-double-cover conjecture asks whether every finite bridgeless graph has at most five Eulerian subgraphs covering each edge exactly twice. It remains unresolved. This note records a structural model and exact computations arising from an attempted proof.

For a loopless cubic graph, a five-cycle double cover is equivalently a conservative labeling of every edge by a two-subset of five coordinates, called a D_5-flow. We show directly that such a flow canonically determines a closed triangulated surface whose faces are dual to the graph vertices and whose primal vertices are properly coloured with five colours. The coordinate circuits are precisely the primal vertex links, and

$$\chi(S(q)) = \sum_{i=0}^{4} \kappa(C_i) - \frac{|V(G)|}{2}.$$

The usual componentwise coordinate transposition is a reversible triangle-band recolouring and regluing surgery. We give an exact permutation formula for its change in Euler characteristic and a cube example in which one switch changes a torus into a sphere.

As a separate delimiter for compressing the new eight-cycle construction, we allow a graph-dependent rule to use the full local affine triangle of an old pair label, requiring only that equal local triangle states reuse the same recolouring. A 12-vertex simple bridgeless cubic witness has connected state-adjacency graphs for all fifteen pairs and eleven distinct triangle rows spanning the binary cycle space of K_6. State transport therefore collapses the rule to a coboundary $r(xy) = a_x + a_y$; weight-two outputs would give a homomorphism $K_6 \to R_5$, impossible because $\omega(R_5) = 5$. Order twelve is sharp for this connected-state/full-rank mechanism. The witness is Tait-colourable, and its positive cover uses different local permutations at two occurrences of the same old state, so this is a precise no-go for state-only compression, not a FiveCDC counterexample.

We also prove a four-coordinate locking lemma and give a separately checkable 14-vertex contraction whose complete generalized-switch orbit has twelve normalized states, none satisfying the boundary condition needed to undo the contraction. This refutes a natural reconfiguration lemma, not the five-cycle-double-cover conjecture. A complementary Tait quotient proves that the analogous fully circuit-trapped branch is root-universal on cubic graphs, explaining why the degree-four trap does not lift unchanged. More generally, a direct Tait-colouring formula realizes any prescribed circuit as exactly one bichromatic factor. Consequently every two edges of a 2-connected Tait-colourable cubic graph share a factor component, and any cubic edge insertion whose eliminated graph is Tait-colourable has an explicit D_5-flow. Thus the unresolved edge-elimination premise is confined to non-Tait graphs. A compact certificate of 30,858 explicit flows, checked without the SAT generator, proves that every simple cyclically 4-edge-connected cubic graph through order 28 is feasible for every independent root pair. The non-Tait part comprises

14,009 canonically generated graphs and 10,689,351 root pairs; the statement is finite and conditional on the documented canonical-generator source boundary. This is an explicit certification of a known stronger finite phenomenon, not a new order frontier: Brinkmann, Goedgebeur, Hägglund, and Markström proved the corresponding strong 5-CDC statement through order 34 [3, Observation 6.6]. We also construct a planar cubic capped-ladder family proving that no constant number of root-component Kempe switches can always rescue two prescribed edges; this concerns a restricted strategy and does not refute rooted Kempe universality or FiveCDC. We prove that universal rooted feasibility on the smaller graph obtained by eliminating one edge of a minimum counterexample would imply FiveCDC; a still weaker sufficient premise only concerns independent roots in 3-edge-connected cubic graphs with the short-cycle geometry forced by edge elimination from girth ten. We give a linear-size exact SAT/XOR certificate for the rooted factor-connectivity condition, using local transition variables whose boundary is the prescribed root pair. For cubic three-edge cuts we also isolate a six-state typed cap signature, prove an internal/external-mode gluing lemma, and exhaustively compute all 138,144 rooted interfaces on biconnected simple cubic graphs through order 14; every computed signature contains a double star. The corresponding universal double-star premise is open. At the same finite orders, every interface also has one individual flow whose external root factors collectively meet all three cap ports; independent enumeration reaches order 12 and a third implementation rechecks order 14. This all-flow property is not orbitwise, and its universal form is open. In the same rooted language, a compact positive certificate and separately written semantic checker cover all 1,301,664 interfaces on the 3,874 biconnected simple cubic graphs of order 16, all interfaces in retained canonical cyclically-four non-Tait streams through order 26, and seven literal strong snarks of order 34: 2,674,404 interfaces in total. The latter rows depend on the documented corpus boundary and do not constitute a complete order-34 theorem. In the marked-girth domain actually forced by edge elimination, a nontrivial three-cut must separate the roots; a self-contained edge-rooted nonbacktracking-walk count shows that each resulting cap has order at least 58 and the restored parent has order at least 116. Two Tait caps always glue by external-mode states. In any cap carrying a D_5-flow, a physical port adjacent to the root is automatically covered by an external state; the proof is a six-case local table. The exact surviving branch therefore has a non-Tait cap of order at least 58 and concerns only ports independent of the root. One 890-vertex non-Tait girth-ten control has the full six-state signature by six explicit checked flows, although one individual flow misses external coverage at one port. Thus the marked-girth blockers do not force externality without changing the flow. Literal certificates also make all 1,335 edge eliminations of this named graph root-good; this remains a one-graph finite result. For separated two- and three-edge cuts we refine the six-state union to an exact finite relation and prove a natural-join formula for rooted external coverage. A literal checker exhausts two complete cut examples and a bounded census finds no bad pair among 24,039 realizable relation pairs through host order ten. We also give a human-checkable weight-four complement translation: on an Eulerian subgraph whose labels omit one coordinate, complementing every label inside the other four coordinates preserves the flow and toggles exactly the four factors containing the omitted coordinate. Explicit translations cross the previously isolated Kempe-orbit barriers. A complete order-14 audit finds 12 interfaces where no single fixed-coordinate Eulerian complement immediately works; every one nevertheless has a disjoint complement five-circuit into a good Kempe orbit. This refutes the one-step strengthening but leaves an exact enlarged-orbit lemma open. Neither the natural join nor this enlarged move proves that every realizable cap pair is compatible. Finally, exhaustive computations show that all 33,598 terminal equal-Euler-characteristic Kempe plateaus on the 480 biconnected simple cubic graphs of order 14 are root-universal. A sharper fixed-distance descent test has 642,167 subplateaus and no failure. These are finite results only. The universal terminal-plateau statement is left explicitly as a conjecture.

1 Status, conventions, and contribution boundary

A k-cycle double cover, in the convention used here, is a list of at most k Eulerian edge-subsets such that every edge occurs in exactly two members. Empty members may be appended, so "at most five" and "five, allowing empty members" are equivalent. This agrees with the convention in Oum's recent exposition: the general cycle-double-cover theorem gives eight Eulerian subgraphs, while the reduction from eight to five remains Conjecture 18 [13]. Recent work on five-cycle double covers also continues to state the five-subgraph assertion as a conjecture and proves special cases [15, 16]. Hušek and Šámal give the current flow-only equivalence and a bijective description of the eight-cycle lifting data [12]. Their Theorem 3.16 and Conjecture 3.19 provide unrooted flow-only context; the linear rooted-transition certificate below is a separate elementary connectivity encoding. We do not claim that encoding as new without a specialist prior-art review.

Finite boundary representations and the gluing of circuit double covers on ordered multipoles are established tools [11]. The exact rooted D_5 relations below specialize that general philosophy; we make no novelty claim for finite-state composition as such. The weight-four translation law and the particular rooted state quotient appear distinct in the sources checked for this draft, but this is a bounded negative search, not proof of priority.

The orientable five-cycle-double-cover conjecture adds directed Eulerian orientations so that each edge is used once in each direction. It is a different, stronger statement; no orientation condition is imposed in this note.

The embedding aspect is established background, not a novelty claim. Catlin explicitly studied embedded graphs, facial colourings, and double cycle covers [5]; see also Zhang's monograph [18] and Mohar and Thomassen's text [17]. Ghanbari and Šámal formulate the cubic CDC in terms of embeddings without singular edges and study both approximation and twist operations [7, 8]. We have not completed a publication-grade prior-art search for the exact D_5 five-colour specialization, the component-switch Euler formula below, or the particular finite trap. Accordingly, this draft makes no categorical novelty claim. Its present value is that the statements, proofs, certificates, and limitations are exposed in a form that can be checked independently.

Literature-search TODO. Before circulation as a preprint, conduct a targeted search for prior work on unbounded Kempe-switch distance in two-edge-cut chains, especially prefix/suffix word obstructions for cycle-double-cover or nowhere-zero flow reconfiguration, and for typed multipole signatures or transition systems equivalent to the rooted certificates below. The capped-ladder theorem and the internal/external cap formalism have not yet been compared adequately against that literature.

2 D_5-flows

All graphs in the structural sections are finite and loopless. Parallel edges are permitted unless simplicity is stated. Edge objects, including parallel ones, are distinct.

Put

$$D_5 = \binom{\{0, 1, 2, 3, 4\}}{2} \subset \mathbb{F}_2^5.$$

We identify a pair ij with its weight-two incidence vector. A D_5-*flow* on a graph G is a map

$$q : E(G) \longrightarrow D_5$$

such that

$$\bigoplus_{e \ni v} q(e) = 0 \tag{1}$$

at every vertex v. Since the graphs considered here are loopless, each incident edge appears once in (1).

Proposition 1 (cover–flow equivalence)**.** *The D_5-flows on G are in bijection with ordered five-tuples $(C_0, \ldots, C_4)$ of Eulerian edge-subsets in which every edge occurs exactly twice.*

Proof. Given q, put

$$C_i = \{e \in E(G) : i \in q(e)\}.$$

The i-th coordinate of (1) says that C_i has even degree at every vertex. Every label has weight two, so every edge belongs to exactly two of the C_i.

Conversely, given the five Eulerian subsets, label an edge by the two indices of the members containing it. The label has weight two, and the five even-degree conditions are exactly (1). The constructions are inverse. □

The following elementary local observation drives the surface model.

Lemma 2 (local coordinate triangle)**.** *Let $a, b, c \in D_5$ and $a + b + c = 0$. Then there are three distinct coordinates i, j, k for which*

$$\{a, b, c\} = \{ij, ik, jk\}.$$

Proof. We have $c = a + b$. Since

$$\mathrm{wt}(a + b) = 4 - 2|a \cap b| = 2,$$

the two pairs a, b meet in exactly one coordinate. Their union has three coordinates, and $c = a \triangle b$ is the third side of that coordinate triangle. □

3 A triangle-state delimiter for eight-to-five compression

Write $W = \mathbb{F}_2^3$. In the eight-cycle construction of Oum and of Hušek–Šámal, every edge receives an unordered pair $P_e \in \binom{W}{2}$, and at a cubic vertex v the three incident pair labels are the sides of a three-point state

$$T_v = \{x, y, z\}, \qquad xy, \quad xz, \quad yz. \tag{C1}$$

Conversely, every closed cubic pair labelling with this local form is compatible with the lifting construction: give xy the flow value $x + y$, and give the state xyz the local potential $x + y + z$. This is the pair form of the lifting construction in [13, 12].

Definition 3 (triangle-state rule)**.** For one fixed supplied pair-labelled cover, a triangle-state rule chooses, after seeing the whole graph,

$$r_T(p) \in D_5 \qquad (T \text{ an occurring state, } p \in \tbinom{T}{2}).$$

It must satisfy local parity

$$\sum_{p \in \binom{T}{2}} r_T(p) = 0 \tag{C2}$$

at every occurring state and edge agreement

$$r_{T_u}(p) = r_{T_v}(p) \tag{C3}$$

when an old edge uv has pair label p. Thus the rule may depend on the complete local affine triangle, but two vertices having the same state must reuse the same three output labels.

For a fixed old pair p, let its *state-adjacency graph* A_p have as vertices the distinct occurring states containing p. Every old edge labelled p joins its two endpoint states; loops and parallel edges may be discarded.

Lemma 4 (state transport)**.** *If A_p is connected, edge agreement forces $r_T(p)$ to have one value $r(p)$, independent of the state T containing p.*

Proof. Equation (C3) is equality along every edge of A_p, so equality propagates along every path. □

Lemma 5 (triangle-cocycle rigidity)**.** *Let S be a set of old coordinates. Suppose all A_p are connected and the incidence rows of the occurring states span $Z_1(K_S; \mathbb{F}_2)$. Every triangle-state rule then has words $a_x \in \mathbb{F}_2^5$, unique up to a common translation, such that*

$$r(xy) = a_x + a_y \tag{C4}$$

for every old pair xy.

Proof. Lemma 4 first makes every value depend only on the old pair. Coordinate by coordinate, equation (C2) says that the resulting edge cochain r is orthogonal to the supplied triangle rows. If those rows span the full cycle space, r belongs to its orthogonal complement, the cut space. Fixing $o \in S$, putting $a_o = 0$ and $a_x = r(ox)$, the triangle equation on oxy gives (C4). Any two such potentials differ by one common word. □

Let R_5 be the graph on $\mathbb{F}_2^5$ joining two words at Hamming distance two. Its clique number is five. Indeed, translate a clique to contain zero. The other words are pairwise-intersecting two-subsets of [5]; they lie in a four-edge star or, once a triangle occurs, in a family of size at most three. Including zero gives at most five, attained by zero and the four pairs of a four-edge star.

Theorem 6 (sharp triangle-state no-go)**.** *There is a 12-vertex simple bridgeless cubic graph with a supplied Oum-compatible eight-cover on six active coordinates for which no triangle-state rule into D_5 satisfies* (C2)–(C3)*. Order twelve is minimum for this connected-state/full-rank certificate.*

Proof. Use the following states in vertex order, with 012 repeated:

$$\begin{matrix} 012 & 012 & 013 & 014 & 023 & 025 \\ 045 & 124 & 125 & 135 & 234 & 345. \end{matrix} \tag{C5}$$

Pair their side occurrences into the eighteen labelled graph edges

$$\begin{matrix} 0{-}2:01 & 1{-}3:01 & 0{-}4:02 & 1{-}5:02 \\ 2{-}4:03 & 3{-}6:04 & 5{-}6:05 & 0{-}7:12 \\ 1{-}8:12 & 2{-}9:13 & 3{-}7:14 & 8{-}9:15 \\ 4{-}10:23 & 7{-}10:24 & 5{-}8:25 & 10{-}11:34 \\ 9{-}11:35 & 6{-}11:45. & & \end{matrix} \tag{C6}$$

The list directly gives a simple connected cubic graph. Deleting each of its eighteen edges leaves it connected, so it is bridgeless; its graph6 record in this order is `KQh?k‘CGGQ?R`. At each vertex the incident labels are the sides of its state in (C5), so the supplied cover is Oum-compatible, with old coordinates 6 and 7 empty.

All fifteen pairs on $\{0,\ldots,5\}$ occur. Twelve occur on one edge between their two states. For the other three, the two edges connect the repeated state 012 respectively to

$$\begin{array}{c|c} 01 & 013, 014 \\ 02 & 023, 025 \\ 12 & 124, 125. \end{array}$$

Thus every A_p is connected. In pair-column order

$$01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 25, 34, 35, 45,$$

the eleven distinct rows in (C5) have binary rank ten. This equals $\dim Z_1(K_6; \mathbb{F}_2) = 15 - 6 + 1$, so Lemma 5 applies. A putative rule would give six words a_x with every difference $a_x + a_y$ of weight two, hence a $K_6 \to R_5$ homomorphism, contradicting $\omega(R_5) = 5$.

For sharpness of this mechanism, a smaller cubic witness could only have order ten: rank ten needs at least ten distinct states and cubic order is even. At order ten the ten rows would be distinct and independent. Full rank forces all fifteen pair columns to occur. Their thirty incidences, together with positive even incidence of every pair in a closed cover, would make every pair occur exactly twice. The xor of all ten triangle rows would then be zero, contradicting independence. Hence order at least twelve is necessary, and (C5)–(C6) attains it. □

The restriction is visible in a positive control. In the edge order (C6),

$$0, 1, 1, 2, 2, 2, 0, 2, 0, 1, 0, 2, 0, 1, 1, 2, 0, 1$$

is a proper 3-edge-colouring; mapping its colours to $01, 02, 12$ gives a FiveCDC. At vertex 0, the sides $01, 02, 12$ of old state 012 receive colours $0, 1, 2$, while at vertex 1 the same sides receive $1, 2, 0$. The successful cover therefore uses different local permutations at two occurrences of the same old state. Allowing arbitrary vertex-specific local rules removes the restriction entirely: compatible vertex-specific output triangles are exactly a FiveCDC labelling. The theorem excludes only the state-only intermediate rule, not bounded-neighbourhood rules or the existential FiveCDC problem.

Configuration homomorphisms and the Desargues/FiveCDC equivalence are prior art [14]. A bounded comparison with that work and the two recent eight-cover sources found no exact state-transport theorem or this witness, but priority remains unestablished. Two independently structured deterministic checkers and the source audit are in the companion triangle-state scratch package dated 20260731.

4 The coloured-surface dictionary

We first give a construction that does not assume an embedding in advance.

Theorem 7 (canonical coloured surface)**.** *Let G be a connected loopless cubic graph with a D_5-flow q. Then q canonically determines, up to colour-preserving cellular isomorphism, a connected closed triangulated surface $S(q)$ with the following properties.*

1. *The triangles of $S(q)$ are indexed by $V(G)$, and the dual graph is G.*
2. *The primal vertices are properly coloured with coordinates $0, \ldots, 4$, in the sense that the three corners of every triangle have distinct colours.*
3. *If $C_i = \{e : i \in q(e)\}$, the components of C_i are in bijection with the primal vertices of colour i.*

Consequently, if $n = |V(G)|$, then

$$\chi(S(q)) = \sum_{i=0}^{4} \kappa(C_i) - \frac{n}{2}. \tag{2}$$

Proof. For each $v \in V(G)$, Lemma 2 gives three coordinate colours on the incident labels. Take an abstract triangle Δ_v with those vertex colours, and assign the side opposite the remaining colour to the incident graph edge whose label is the two endpoint colours of that side.

For every graph edge $e = uv$, glue the side of Δ_u indexed by e to the side of Δ_v indexed by e, using the unique map that preserves its two endpoint colours. Every triangle side is paired exactly once. The quotient is therefore a closed two-dimensional pseudomanifold.

A quotient vertex can initially have more than one link component. Replace it by one vertex for each link component. Each resulting vertex has a circular link, so the normalized quotient is a closed triangulated surface. Splitting vertices does not change the triangles, their side pairings, or the dual graph. The surface is connected because its dual graph G is connected.

Fix a coordinate i. In a triangle having an i-coloured corner, the two sides at that corner are exactly the two incident graph edges whose labels contain i. Following the link of that corner through the side pairings therefore follows a component of C_i, and every such component arises this way. This proves (1)–(3).

The surface has n triangles and $3n/2$ primal edges. Its number of vertices is $\sum_i \kappa(C_i)$. Euler's formula now gives

$$\chi(S(q)) = \sum_i \kappa(C_i) - \frac{3n}{2} + n,$$

which is (2). □

For a precise converse, call a face-colouring of a cellular embedding of a cubic graph *corner-proper* when the three face incidences at each graph vertex have three distinct colours. This definition rules out a single face occupying two corners at one vertex.

Proposition 8 (converse)**.** *A corner-proper five-colouring of the faces of a cellular embedding of a connected cubic graph gives a D_5-flow: label an edge by the colours of its two incident face occurrences. This construction and Theorem 7 are inverse after normalizing quotient vertices by link components.*

Proof. At a cubic vertex, let the three incident face colours be i, j, k. The three incident edge labels are ij, ik, jk, whose xor is zero. Thus the labels form a D_5-flow. Taking the dual triangle at every graph vertex recovers the original side pairings and face-colour occurrences. Normalization separates exactly the distinct link components represented by distinct facial occurrences. □

Remark 9. Propositions 1 and 8 show that the standard, unoriented five-cycle-double-cover problem for cubic graphs can be read as a problem of finding a corner-proper five-face-coloured cellular embedding. This is not an orientability statement: $S(q)$ may be orientable or nonorientable.

5 Bichromatic boundary curves and Kempe surgery

For $i \neq j$, define

$$Y_{ij} = C_i \triangle C_j = \{e : |q(e) \cap \{i, j\}| = 1\}. \tag{3}$$

It is an even subgraph and, in a cubic graph, a disjoint union of circuits.

Proposition 10 (boundary interpretation)**.** *In $S(q)$, let H_{ij} be the primal subgraph induced by the primal vertices of colours i, j. After the standard dual isotopy,*

$$\partial N(H_{ij}) = Y_{ij}, \tag{4}$$

where $N(H_{ij})$ is a sufficiently small regular neighbourhood.

Proof. Check one primal triangle. If it contains neither i nor j, the neighbourhood has no boundary arc there. If it contains exactly one of them, the boundary arc around that corner crosses the two triangle sides whose labels contain exactly one of i, j. If it contains both, the boundary arc around their common primal edge crosses the other two sides, again exactly those whose labels contain one of i, j. These local arcs join across paired triangle sides and give (4). □

Let K be a circuit component of Y_{ij}. A *component Kempe switch* transposes i, j in the label of every edge of K.

Lemma 11 (validity of a component switch)**.** *A component Kempe switch sends a D_5-flow to a D_5-flow and is an involution.*

Proof. For an active label, transposing i, j adds the vector ij. At each graph vertex the circuit K has degree zero or two, so the total added vector in the local xor equation is zero. The switched labels still have weight two. Applying the same transposition a second time restores the original state. □

On the coloured surface, this move has a concrete description. Recolour the triangles traversed by K by the transposition $i \leftrightarrow j$. Active side gluings still match because both incident triangles are recoloured. At an inactive ij-side incident with exactly one recoloured triangle, compose the side gluing with the endpoint flip. The result is exactly the coloured quotient for the switched flow. Thus the move is a reversible triangle-band recolouring and regluing surgery. It need not act on one fixed topological surface.

5.1 An exact Euler-change formula

Fix i, j, and abbreviate $A = C_i$, $B = C_j$. At each vertex of $A \cup B$, the nonzero membership patterns in (A, B) are

$$10, \qquad 01, \qquad 11.$$

Discard components of $A \cup B$ that are pure circuits. In every remaining component suppress the degree-two paths. The result is a cubic multigraph T_{ij} whose three edge classes $10, 01, 11$ are perfect matchings. Write m_{10}, m_{01}, m_{11} for the associated fixed-point-free involutions on $V(T_{ij})$.

For a permutation π, let $z(\pi)$ denote its number of cycles, including fixed points.

Theorem 12 (one-switch Euler formula)**.** *Let K be a component of Y_{ij} meeting T_{ij}, and let W be the vertices on the corresponding alternating circuit of $m_{10} \cup m_{01}$. Define*

$$\begin{array}{c|cc} & W & V(T_{ij}) \setminus W \\ \hline m'_{10} & m_{01} & m_{10} \\ m'_{01} & m_{10} & m_{01}, \end{array} \qquad m'_{11} = m_{11}. \tag{5}$$

If q' is obtained by switching on K, then

$$\begin{aligned} \chi(S(q')) - \chi(S(q)) = \frac{1}{2}\big[&z(m'_{10}m_{11}) + z(m'_{01}m_{11}) \\ &- z(m_{10}m_{11}) - z(m_{01}m_{11})\big]. \end{aligned} \tag{6}$$

If K is a discarded pure circuit, the Euler change is zero.

Proof. Only $C_i = A$ and $C_j = B$ can change, so by (2) the Euler change is

$$\Delta\kappa(A) + \Delta\kappa(B). \tag{7}$$

The components of A that meet the cubic core correspond to the alternating circuits of the two matchings m_{10}, m_{11}; similarly, the core components of B correspond to the alternating circuits of m_{01}, m_{11}.

If m_a, m_b are two perfect matchings, then on each alternating circuit of their union the product $m_a m_b$ has two permutation cycles. Therefore the number of alternating circuits is $\frac{1}{2}z(m_a m_b)$.

Switching i, j on K exchanges the 10- and 01-matching edges at exactly the vertices of W, giving (5). Substitute the before-and-after alternating-circuit counts into (7) to obtain (6). Pure circuit components outside the cubic core are unchanged or move whole from A to B; their total contribution to (7) cancels. □

5.2 A hand-checkable topology change

Take the cube with edge order

e	0	1	2	3	4	5	6	7	8	9	10	11
uv	01	03	04	12	17	23	26	35	45	47	56	67

and labels

$$01, 02, 12, 02, 12, 01, 12, 12, 01, 02, 02, 01. \tag{8}$$

The reader can check the xor equation at each of the eight vertices directly from the table. The coordinate component counts are

$$(\kappa(C_0), \ldots, \kappa(C_4)) = (2, 1, 1, 0, 0),$$

so (2) gives $\chi = 0$. The edges $\{0, 1, 3, 5\}$ form one Y_{12}-circuit. Transposing $1, 2$ on those four edges changes the component counts to

$$(2, 2, 2, 0, 0),$$

so the new connected surface has $\chi = 2$. A triangle-orientation propagation check shows that both surfaces are orientable. Hence this single legal switch changes a torus into a sphere. In particular, a global proof cannot treat all Kempe moves as curve slides on one fixed surface.

6 A four-coordinate locking mechanism

It is useful to distinguish component switches from a stronger move. For a coordinate pair $p = ij$, a *generalized factor switch* chooses any Eulerian edge-subset $Z \subseteq Y_{ij}$ and adds the vector ij to every label on Z. This remains a D_5-flow by the same parity argument as Lemma 11.

Lemma 13 (four-coordinate one-step lock)**.** *Suppose a D_5-flow uses exactly the four coordinates $S = \{0, 1, 2, 3, 4\} \setminus \{t\}$, and suppose every nonempty coordinate class C_i, $i \in S$, is one circuit. From this state, no generalized factor switch can introduce t nontrivially modulo a global coordinate permutation.*

Proof. A switch with pair $p \subseteq S$ does not introduce t. For a pair $p = \{i, t\}$, the absence of t gives

$$Y_{it} = C_i.$$

A circuit has only two Eulerian edge-subsets, the empty set and the whole circuit. The empty switch does nothing. Switching all of C_i replaces i by t on every edge label containing i, which is exactly the global coordinate transposition $i \leftrightarrow t$. □

The hypothesis of Lemma 13 need not be preserved by all switches among the used coordinates. Thus the lemma alone is a one-step statement, not an orbit theorem. The following finite example separately certifies orbit closure.

Proposition 14 (certified generalized-switch trap)**.** *There is a simple 3-connected cubic graph G on 14 vertices and an edge e such that, after contracting e, the resulting degree-four graph has a generalized-factor-switch orbit with the following properties, modulo the global S_5 action:*

1. *the orbit has exactly 12 states;*
2. *every state uses four coordinates and each used coordinate class is one circuit; and*
3. *every state has side xor of weight four and therefore cannot be extended over the restored edge.*

The same contracted graph also has a different orbit containing extendible states. Therefore this is not a counterexample to FiveCDC.

Finite certificate. The parent graph has graph6 string

`M?AA@BORD_CoEOAo?.`

In lexicographic edge order, contract edge $15 = (3, 12)$. With the contracted edges indexed by the inherited order, the prescribed sides at the degree-four vertex are

$$U = \{5, 11\}, \qquad V = \{15, 16\}.$$

One state in the orbit is

$$\begin{aligned}&(01, 02, 03, 13, 03, 01, 23, 02, 03, 03, 02, 23, 23, 13, 12,\\ &\ 01, 23, 12, 01, 02).\end{aligned} \tag{9}$$

Its local xors vanish, while both side xors are

$$q(5) + q(11) = q(15) + q(16) = 0123. \tag{10}$$

The companion file `orbit-certificate.txt` lists all twelve normalized rows and the complete normalized adjacency relation. A finite hand verification proceeds as follows. For each row, check the local xor equations and (10). For each coordinate pair, construct Y_{ij}, delete one edge from every circuit to obtain a spanning forest, and use the remaining edges as a binary cycle-space basis. Switch every basis sum and normalize under the 120 coordinate permutations. The resulting distinct rows are exactly the neighbours listed in the certificate. The adjacency list is connected, so the twelve rows form one complete orbit. Each row also directly satisfies the hypotheses of Lemma 13.

The supplied independent Python checker performs exactly these finite operations without a SAT solver, graph package, or canonical-labelling library. It also checks cubicity, bridgelessness, and 3-connectivity of the parent. □

The proposition refutes the tempting assertion that every starting flow on a contraction can be repaired by generalized factor switches. It does not refute the existential assertion that some flow on the contraction extends; in fact, the other orbit contains 968 normalized extendible states.

6.1 Why the degree-four trap does not lift unchanged

The contraction trap raises a natural concern: perhaps a cubic Kempe orbit could remain locked in four coordinates with all four coordinate classes connected and still fail rooted universality. The next theorem excludes exactly that branch.

Call a component-Kempe orbit on a cubic graph *four-coordinate circuit-trapped* when every state uses exactly four coordinates and every used coordinate class is one circuit.

Theorem 15 (cubic D_4 trap theorem)**.** *Every four-coordinate circuit-trapped component-Kempe orbit on a cubic graph is root-universal. More precisely, every pair of root edges is rescued in the starting state or after one component switch between used coordinates.*

Proof. Fix a state q, write S for its four used coordinates and t for the missing coordinate, and let r, s be distinct root edges. For clarity, globally rename S as $\{0, 1, 2, 3\}$.

Identify complementary labels in the three classes

$$\{01, 23\}, \qquad \{02, 13\}, \qquad \{03, 12\}. \tag{11}$$

These are three Tait colours. At each cubic vertex, the local xor triple is the triangle on three of the four coordinates, so its three labels lie in distinct classes in (11). Thus the quotient is a proper three-edge-colouring.

If $q(r) \cap q(s)$ contains a coordinate i, then both roots lie in

$$C_i = Y_{it}.$$

The circuit-trap hypothesis says that C_i is one circuit, so the roots are already rescued.

Otherwise the two root labels are disjoint. They are complementary two-subsets of S, hence have the same Tait colour A. Choose either other Tait colour B. The union of classes $A \cup B$ is a factor Y_p, where $\{p, S \setminus p\}$ is the third class. If r, s lie on the same component of Y_p, they are already rescued.

If they lie on different components, transpose the two coordinates of p on the component containing r. In the Tait quotient this interchanges the two active colours on that component: the colour of r changes from A to B, while the colour of s remains A. Labels from distinct complementary classes in (11) meet in exactly one coordinate, say k. The literal switch did not introduce t, so both roots now lie in $C'_k = Y'_{kt}$. The switched state remains in the assumed circuit-trapped orbit, and therefore C'_k is one circuit containing both roots. □

The proof uses cubicity to obtain a proper Tait quotient and alternating bichromatic factor circuits. Neither fact survives at the degree-four contraction vertex. Theorem 15 therefore explains why Proposition 14 cannot be transplanted unchanged to a cubic counterexample. It closes only the fully circuit-trapped four-coordinate branch; a general rooted-orbit obstruction could still use five coordinates, a disconnected coordinate class, or transitions between those regimes.

7 Prescribing a factor circuit from a Tait colouring

The preceding quotient argument has a stronger constructive form that does not assume a circuit-trapped orbit. Its existence conclusion is not new: Hoffmann-Ostenhof records the stronger fact that every 2-regular subgraph of a subdivision of a cubic graph with a nowhere-zero 4-flow belongs to a 4-CDC [9, Lemma 0.2]. Since a Tait-colourable cubic graph has such a flow, the theorem below is a direct explicit D_5-formula for that known case, followed by the root and insertion adaptations needed here. Recall that, for a coordinate pair P, the factor

$$Y_P(q) = \{e : |q(e) \cap P| = 1\}$$

is Eulerian.

Theorem 16 (prescribed Tait factor)**.** *Let H be a loopless cubic graph with a proper three-edge-colouring, and let K be any circuit of H. There is a D_5-flow q such that*

$$Y_{01}(q) = E(K).$$

Proof. Name the three Tait colours $2, 3, 4$. If an edge e has colour c, put

$$q(e) = \begin{cases} 0c, & e \in E(K), \\ \{2,3,4\} \setminus \{c\}, & e \notin E(K). \end{cases} \tag{T}$$

Every label has weight two. At a vertex outside K, the three incident labels are $23, 24, 34$, whose xor is zero. At a vertex of K, the two circuit edges have distinct colours a, b, while the third edge has the remaining colour c. The three labels in (T) are

$$0a, \quad 0b, \quad ab,$$

again with xor zero. Thus q is a D_5-flow. Coordinate 1 occurs nowhere, and a label meets 01 oddly exactly when it contains coordinate 0. By (T), those edges are exactly $E(K)$. □

Corollary 17 (Tait root universality)**.** *If H is a 2-connected Tait-colourable cubic graph, then every two distinct edges r, s belong to one factor component of some D_5-flow.*

Proof. Every two edges of a 2-connected graph lie on a common circuit. Indeed, subdivide both edges. Subdivision preserves 2-connectivity, and the vertex form of Menger's theorem gives a circuit through the two new vertices. Suppressing them gives a circuit K of H containing r, s. Theorem 16 makes K exactly the nonempty factor Y_{01}. □

Corollary 18 (Tait edge insertion)**.** *Let G be obtained from a 2-connected Tait-colourable cubic graph H by subdividing two distinct edges r, s with vertices u, v and adding the edge uv. Then G has a D_5-flow, and hence a five-even- subgraph double cover.*

Proof. Choose a circuit K through r, s, apply Theorem 16, and subdivide r, s, giving both halves their old labels. The subdivided K is a circuit K^* through u, v. Choose either u-to-v arc A of K^*, transpose coordinates $0, 1$ on every edge of A, and label the new edge uv by 01.

At an internal vertex of A, two incident labels are transposed, so xor conservation remains true. At each of u, v, exactly one old label changes by xor with 01, and the new label 01 cancels that defect. Every transposed label was $0c$ and becomes $1c$, so it still has weight two. The resulting labels are therefore a D_5-flow on G. Proposition 1 gives the asserted cover. □

Thus a reduced graph produced by the two-root edge-elimination induction cannot obstruct extension merely by being Tait-colourable. The remaining rooted premise is genuinely confined to non-Tait reduced graphs. This is a reduction of that premise and an explicit adaptation of a known prescribed-subgraph result, not a resolution of FiveCDC.

7.1 An explicit rooted certification through order 28

Call a cubic graph *independently root-universal* if, for every two vertex-disjoint edges r, s, it has a D_5-flow in which some factor component contains both roots.

Theorem 19 (explicit finite certification)**.** *Every simple cyclically 4-edge-connected cubic graph of order at most 28 is independently root-universal, conditional on the documented completeness and option semantics of the retained Snarkhunter 2.0b graph streams.*

The order cutoff in Theorem 19 is not new. Brinkmann, Goedgebeur, Hägglund, and Markström proved that every circuit in every bridgeless cubic graph through order 34 belongs to a 5-CDC [3, Observation 6.6]. Apply the standard two-edge insertion to independent roots of a graph of order at most 32. The inserted graph is bridgeless, cubic, and has order at most 34, so the published observation supplies a 5-CDC; Lemma 25 then supplies the rooted flow. Thus the existential statement below order 28 is strictly subsumed by prior work. The point of the theorem here is the literal specialized certificate and its solver-independent checker.

Proof. A cyclically 4-edge-connected cubic graph is 2-connected: a bridge shore in a cubic graph necessarily contains a circuit. Thus the Tait-colourable case is Corollary 17. The Petersen graph is the smallest bridgeless non-Tait cubic graph, so for the remaining case the retained canonical streams begin at order ten and contain

$\lvert V\rvert$	10	12	14	16	18	20	22	24	26	28
graphs	1	0	0	0	2	6	31	155	1297	12517.

The generator logs, graph6 hashes, distinct-row counts, and an independent classification are part of the source package. Thus there are 14,009 non-Tait graphs. A simple cubic graph of order n has $\binom{3n/2}{2} - 3n$ independent unordered edge pairs, giving 10,689,351 pairs in total.

The positive certificate records 30,858 literal D_5-flows. A separately written standard-library checker reparses each graph6 row, verifies that every edge label has weight two, xors the three labels at every vertex, and constructs all ten factors Y_P. It then enumerates all independent edge pairs and verifies that each pair occurs in one component of one factor of at least one displayed flow. It recomputes the totals 14,009, 10,689,351, and 30,858. These finite checks cover every non-Tait row, while the first sentence covers every remaining graph. □

The certificate and checker are in `scratch/d5-root-pair-census-20260731`. The theorem neither bounds the order of a possible rooted obstruction nor proves the universal rooted premise. It is not claimed as a novel finite frontier.

8 Unbounded delay for root-component rescue

Fix distinct root edges r, s. Call a state *root-good* when some component of some Y_{ij} contains both roots. While the state is not root-good, a *root-component switch* means a component Kempe switch whose switched circuit contains exactly one root.

For $m \geq 1$, define a capped ladder L_m as follows. Its left cap has vertices a, b, p, q, its ladder has vertices u_i, v_i for $0 \leq i \leq m$, and its right cap has vertices c, d, x, y. The edge set is

$$\begin{aligned}
&ap, aq, bp, bq, pq, \quad au_0, bv_0,\\
&u_iv_i \quad (0 \leq i \leq m),\\
&u_iu_{i+1}, v_iv_{i+1} \quad (0 \leq i < m),\\
&u_mc, v_md, \quad cx, cy, dx, dy, xy.
\end{aligned}$$

Thus the caps are copies of $K_4 - e$, and

$$|V(L_m)| = 2m + 10, \qquad |E(L_m)| = 3m + 15.$$

The graphs are simple planar cubic graphs. They are bridgeless: cap edges lie on cap triangles, ladder edges lie on squares, and the attachment edges lie on cycles running through both rails and the caps.

Put

$$A = 01, \qquad B = 02, \qquad C = 12.$$

There is a Tait-supported D_5-flow on L_m for which the labels on the successive two-edge cuts from left to right form the length-$(m + 2)$ prefix

$$A, C, B, A, C, B, \ldots.$$

Explicitly, label au_0, bv_0 by A, label u_0v_0 by B, and assign the rail-pair and rung labels successively so that every ladder vertex sees A, B, C. On the left cap take

$$q(ap) = q(bq) = B, \quad q(aq) = q(bp) = C, \quad q(pq) = A,$$

and use the unique compatible Tait colouring on the right cap, up to exchanging its two inner vertices. Every vertex then sees A, B, C, so the local xor equations hold.

Theorem 20 (no constant root-component radius)**.** *Choose one root edge inside each cap of L_m in the state above. Every root-component-switch sequence that reaches a root-good state has length at least*

$$\left\lfloor \frac{m+2}{3} \right\rfloor.$$

Consequently, no universal constant bounds the number of root-component switches required for rescue.

Proof. Let $D_0, \ldots, D_{m+1}$ be the nested two-edge cuts in the displayed word. For an arbitrary D_5-flow, the two edges of each D_j have the same label: xor the vertex equations on either side of the cut; internal edges cancel twice, leaving the xor of the two cut labels. Write the resulting cut-label word as $\lambda_0\lambda_1 \cdots \lambda_{m+1}$.

Let K be a factor component containing exactly one root. Connectedness and nestedness imply that the cuts met by K form a prefix when K contains the left root and a suffix when it contains the right root. Moreover, K is Eulerian, so it meets a two-edge cut in zero or two edges. A switch on K therefore applies one coordinate transposition to a prefix or suffix of the cut-label word.

After t switches, their at most t boundaries divide the initial word into at most $t+1$ intervals. On each interval the final labels are obtained from the initial labels by one fixed coordinate permutation, namely the ordered composition of the switches covering that interval.

If the final state is root-good through Y_h, its component containing both roots crosses every D_j. Hence

$$h \cdot \lambda_j = 1 \qquad (0 \le j \le m + 1),$$

with dot product over $\mathbb{F}_2$.

Partition the initial word into $\lfloor (m + 2)/3 \rfloor$ disjoint consecutive triples A, C, B. No one triple can remain in a single interval. Otherwise one coordinate permutation π would act on all three entries, and

$$\pi(A) + \pi(C) + \pi(B) = \pi(A + C + B) = 0.$$

Dotting with h contradicts the preceding display, because the xor of three ones is one, not zero. Thus every disjoint triple contains a switch boundary in one of its two internal gaps. Those gaps are disjoint between triples, so one boundary splits at most one triple. Therefore

$$t \geq \left\lfloor \frac{m+2}{3} \right\rfloor,$$

as required. □

Remark 21 (scope). The proof deliberately relaxes the actual move system and uses only its necessary prefix/suffix action. It proves unbounded distance for the restricted root-component strategy. It does not produce a state that is unrescuable by root-component switches, does not refute unrestricted rooted Kempe universality, and does not prove or disprove FiveCDC. Literal breadth-first searches find finite rescue distances $3, 4, 5, 6, 8, 9$ for selected roots in $L_1, \ldots, L_6$, respectively. These computations are supplementary to, not inputs to, the proof.

9 Surface-Euler plateaus

Let $\mathcal{K}(G)$ be the state graph whose vertices are D_5-flows modulo global coordinate permutations and whose edges are component Kempe switches. Put

$$\Phi(q) = \chi(S(q)).$$

For fixed graph edges r, s, call q (r,s)-*good* if some Y_{ij}-component contains both roots.

Definition 22. A Φ-plateau is a connected component after retaining only the equal-Φ edges of $\mathcal{K}(G)$. It is *terminal* when no state in it has a neighbour of larger Φ. A set of states is *root-universal* when, for every unordered edge pair r, s, it contains an (r,s)-good state.

Lemma 23 (connected-coordinate non-descent)**.** *Suppose every C_i is one nonempty circuit. Then every component Kempe switch $q \mapsto q'$ satisfies*

$$\Phi(q') \geq \Phi(q).$$

If q lies in a terminal plateau, every such first switch is Φ-neutral.

Proof. Only the two switched coordinate classes, say C_i, C_j, can change. Initially their total component count is two. If both resulting classes are nonempty, each has at least one component and their total remains at least two.

Suppose instead that C_i' is empty. Then the switched component contains all of C_i, and $C_i \cap C_j = \varnothing$, since intersection edges are not in Y_{ij}. The two original circuits are also vertex-disjoint: two edge-disjoint even subgraphs of a cubic graph cannot both have degree two at the same vertex. The new C_j' therefore contains the two circuits as distinct components. Its component count is at least two, while C_i' contributes zero. The case $C_j' = \varnothing$ is symmetric. Thus

$$\kappa(C_i') + \kappa(C_j') \geq \kappa(C_i) + \kappa(C_j) = 2,$$

and (2) proves non-descent. Terminality excludes a strict increase. □

Lemma 24. *Every state has a Φ-nondecreasing Kempe path to a terminal plateau.*

Proof. Move freely inside the current equal-Φ plateau. If it is not terminal, reach a state having a neighbour of strictly larger Φ and take that edge. The state graph is finite, so Φ can increase only finitely often. The process ends at a terminal plateau. $\square$

The computationally suggested statement is deliberately isolated.

> **Terminal-plateau conjecture.** For every connected bridgeless cubic graph with a D_5-flow, every terminal Φ-plateau is root-universal.

This statement is unproved. It is stronger than the finite observations below and is not asserted as a theorem.

9.1 Why the terminal conjecture would imply FiveCDC

The preceding conjecture is not merely a reconfiguration statement about graphs already known to have a D_5-flow. Together with the standard minimum-counterexample reductions, it would prove FiveCDC.

Lemma 25 (two-root insertion equivalence)**.** *Let H be cubic, let $r = u_1u_2$ and $s = v_1v_2$ be distinct edges, and obtain G by subdividing r, s with new vertices u, v and adding $e = uv$. Then H has an (r, s)-good D_5-flow if and only if G has a D_5-flow q in which u, v lie on the same component of $Y_{q(e)}(q)$.*

Proof. Choose a component K of $Y_{ij}(q)$ containing both roots. After subdividing r, s, give both halves their old labels, so K becomes a circuit K^* through u, v. Choose either u-to-v arc of K^* and transpose i, j on its edges. Every internal vertex sees either zero or two changed labels. At each of u, v, exactly one label changes by the vector ij. Label the new edge uv by ij; this restores xor zero at both new vertices. All labels still have weight two. The new edge has label ij, so it is inactive in Y_{ij}; transposition preserves activity, and hence the subdivided K remains a factor circuit through u, v.

Conversely, let q be a flow on G, put $P = q(e)$, and let K be the common $Y_P(q)$-circuit through u, v. The edge e is inactive. At u, the other two labels therefore have the form $a, a + P$, and the same holds at v. Transpose the two coordinates of P on one u-to-v arc of K, and delete e. Internal vertex xors are unchanged, while the defect P at each endpoint cancels the deleted label. Exactly one of $a, a + P$ is transposed at each endpoint, so the two surviving labels become equal. Suppressing u, v therefore gives a D_5-flow on H, and the suppressed image of K is one Y_P-circuit containing r, s. $\square$

Lemma 26 (bridgeless elimination)**.** *Let G be a simple cyclically 4-edge-connected cubic graph of girth at least five. For $e = uv$, write*

$$N(u) - v = \{u_1, u_2\}, \qquad N(v) - u = \{v_1, v_2\}.$$

Deleting u, v and adding $r = u_1u_2$, $s = v_1v_2$ produces a smaller simple 3-edge-connected cubic graph $G \div e$.

Proof. The elementary edge-reduction criterion stated immediately before Lemma 2.1 of [4] says that an edge of a connected cubic graph is irreducible only if it is a bridge, has an endpoint in a triangle not containing it, or has both endpoints on a 4-cycle not containing it. Thus the girth and connectivity hypotheses make $G \div e$ connected. The four neighbours displayed above are distinct, neither new edge is a loop or duplicates an old edge, and the two new edges are distinct; otherwise G has a triangle or a 4-cycle. Hence $G \div e$ is simple.

Suppose it has a bridge b, with shores X, Y. If $b = r$, then the endpoints of s lie on one shore while u_1, u_2 lie on opposite shores. Restoring u, v, e leaves the edge from u into the other shore as a bridge of G, a contradiction. The case $b = s$ is symmetric.

Otherwise b is an old edge. Neither r nor s crosses (X, Y), so each of the pairs $\{u_1, u_2\}$, $\{v_1, v_2\}$ lies within one shore. If the two pairs lie on the same shore, b remains a bridge after restoring u, v, e. If they lie on opposite shores, $\{b, e\}$ is a 2-edge-cut of G. Both conclusions contradict 3-edge-connectivity.

It remains to exclude a 2-edge-cut $\delta_H(X)$ in $H = G \div e$. Let t be the number of roots crossing it. If $t = 2$, put the reinserted vertices u, v on the same shore: one spoke from each replaced root crosses and uv does not, giving a 2-edge-cut of G. If $t = 1$, put the vertex belonging to the noncrossing root on that root's shore and put the other new vertex there as well. One old cut edge and one spoke cross, again giving a 2-edge-cut of G. If $t = 0$ and both root pairs lie on one shore, put both new vertices there; the old 2-edge-cut persists. The only remaining placement has the two root pairs on opposite shores. Placing u, v on their respective shores then makes the two old cut edges together with uv a 3-edge-cut of G. Both shores contain circuits: a tree shore X of a 2-cut in a cubic graph would satisfy $3|X| = 2(|X| - 1) + 2$, which is impossible. This contradicts cyclic 4-edge-connectivity. Hence H is 3-edge-connected. □

Proposition 27 (fixed-five minimum reduction)**.** *If the standard Five-Cycle Double Cover Conjecture is false, it has a minimum counterexample that is simple, cubic, cyclically* 4*-edge-connected, and not 3-edge-colourable.*

Proof. Choose a counterexample with as few edges as possible. Components may be covered separately, padding each list with empty members and taking unions member by member. A loop may be deleted and then inserted into any two members, since a loop contributes degree two at its endpoint. Thus the counterexample is connected and loopless.

The usual minimum-counterexample proof for cycle double covers [18] preserves the fixed bound five. For completeness, here are the two reconstructions where that point matters. Across a 2-edge-cut, close the two shores by one artificial edge each. The two artificial edges receive labels in D_5; a global permutation of the five coordinates makes those labels equal, after which deleting the artificial edges and restoring the cut edges pastes the two D_5-flows member by member. Hence a minimum counterexample is 3-edge-connected.

If a vertex has degree at least four, Fleischner's splitting lemma [6, Lemma III.26] supplies two incident edges whose split-off graph remains bridgeless. In a five-cover of that smaller graph, replace every occurrence of the artificial edge by the corresponding two-edge path through the split vertex. This changes no member index and preserves even degree. Minimality therefore makes the counterexample have maximum degree at most three. A degree-two vertex may likewise be suppressed: replace its two incident edges by one edge, then lift a cover by replacing every occurrence of the new edge by the original two-edge path. A connected loopless bridgeless graph has no degree-one vertex, so the minimum counterexample is cubic.

If a nontrivial 3-edge-cut remains, contract each shore in turn to a cubic vertex and cover the two smaller bridgeless graphs. At a cubic vertex the three D_5-labels with xor zero are the three edges of a triangle on three coordinates. Every bijection between two ordered label triangles is induced by a permutation of the five coordinates. After applying such a permutation to one shore, the three boundary labels match and the covers paste without adding a member. Thus no nontrivial 3-edge-cut remains, which for a cubic graph is cyclic 4-edge-connectivity.

A proper 3-edge-colouring with colour classes M_1, M_2, M_3 gives the three-member double cover

$$M_1 \cup M_2, \qquad M_1 \cup M_3, \qquad M_2 \cup M_3;$$

append two empty members. The minimum counterexample is therefore not 3-edge-colourable. A loopless 3-edge-connected cubic multigraph with parallel edges is either separated by a 2-edge-cut or is the two-vertex graph with three parallel edges. The latter is 3-edge-colourable, so the minimum counterexample is simple. □

Proposition 28 (conditional resolution)**.** *The terminal-plateau conjecture implies the standard Five-Cycle Double Cover Conjecture.*

Proof. If FiveCDC is false, Proposition 27 gives a simple cyclically 4-edge-connected cubic minimum counterexample G. Huck proved that such a minimum FiveCDC counterexample has girth at least ten [10]. Choose any edge e. Lemma 26 gives a smaller bridgeless cubic graph $H = G \div e$. By minimality, H has a D_5-flow.

Lemma 24 reaches a terminal plateau in its Kempe orbit. By the terminal-plateau conjecture, that plateau contains a state in which the two new edges r, s of H lie on one factor component. Lemma 25 extends this state over the inverse edge insertion, giving a D_5-flow on G, a contradiction. □

Remark 29 (a weaker sufficient premise). The proof needs neither every Kempe orbit nor a prescribed starting flow. It is enough to prove the following independent-pair feasibility statement:

> Every simple 3-edge-connected cubic graph H with independent edges r, s, having girth at least eight, every 8-circuit through both roots, and every 9-circuit through at least one root, has the following property: if H has a D_5-flow, then it has one in which one factor component contains both roots.

Indeed, for $H = G \div e$ the new edges r, s are vertex-disjoint. A circuit of H using neither, one, or both new edges lifts to a circuit of G of the same length, one greater, or two greater, respectively. Thus $g(G) \geq 10$ implies $g(H) \geq 8$; moreover, every 8-circuit uses both roots and every 9-circuit uses at least one. Lemma 26 gives 3-edge-connectivity, and Lemma 25 finishes the same contradiction. This feasibility statement is still open.

9.2 A linear rooted-transition certificate

Let H be a loopless cubic graph with m edges and let r, s be distinct root edges. For each edge e and coordinate $i \in \{0, \ldots, 4\}$, introduce a Boolean variable $x_{e,i}$. Impose

$$\sum_{i=0}^{4} x_{e,i} = 2, \qquad \bigoplus_{e \ni v} x_{e,i} = 0 \tag{12}$$

for every edge e, vertex v, and coordinate i. These are exactly the D_5-flow constraints of Proposition 1. Fix the factor Y_{01} and put

$$y_e = x_{e,0} \oplus x_{e,1}. \tag{13}$$

At each cubic vertex v, and for each unordered pair $\{e, f\}$ of its three incident edges, introduce $z_{v;\{e,f\}}$ with implications

$$z_{v;\{e,f\}} \Longrightarrow y_e, \qquad z_{v;\{e,f\}} \Longrightarrow y_f. \tag{14}$$

Finally, for every graph edge e, impose the transition-boundary xor

$$\bigoplus_{\substack{v \text{ endpoint of } e \\ f \in E(v)\setminus\{e\}}} z_{v;\{e,f\}} = \begin{cases} 1, & e \in \{r, s\}, \\ 0, & e \notin \{r, s\}. \end{cases} \tag{15}$$

Proposition 30 (linear rooted-transition characterization). *Constraints* (12)–(15) *are satisfiable if and only if H has a D_5-flow in which r, s lie on one circuit component of Y_{01}. In a cubic graph with m edges, the direct encoding has $8m$ variables. Using native pseudo-Boolean exact-cardinality constraints and native xor is linear in m; the explicit CNF conversion below has $133m/3$ clauses.*

Proof. The xor of the two coordinate equations at a vertex makes the active y-degree zero or two. Form the transition graph $T(Y_{01})$: its vertices are the active graph edges, and at every active graph vertex join the two active incident edge-nodes. It is a disjoint union of circuits, with components exactly the circuit components of Y_{01}.

If r, s lie on one such circuit, select the z-transitions of either r-to-s path in that component. Their edge-node boundary is exactly $\{r, s\}$, so (14)–(15) hold. Conversely, (14) makes the selected z's an edge subset of $T(Y_{01})$. Every connected component of an edge set has an even number of odd-degree vertices. Equations (15) make r, s the only odd-degree vertices, so they lie in one selected component and hence in one circuit of $T(Y_{01})$. No explicit at-most-one constraint on the three z's at a graph vertex is missing: (12)–(13) allow only zero or two active incident edges, so at most one local pair can satisfy both implications.

For the size count, exact weight two among five variables uses ten negative triple clauses and five positive four-variable clauses, for $15m$. A three-variable even-parity equation uses four clauses, contributing $20n$ for the five coordinate equations at the $n = 2m/3$ vertices and $4m$ for (13). The six implications per vertex contribute $6n = 4m$. Each edge-node occurs in four possible local transitions, so its boundary xor has eight clauses, contributing $8m$. The total is $15m + 20n + 4m + 4m + 8m = 133m/3$. The variables number $5m + m + 3n = 8m$. ☐

The formulation uses edge identities, so parallel edges need no change. Loops are excluded: a loop contributes twice, hence zero, to vertex parity, and would require doubled incidence in both (12) and (15). A satisfying assignment is a directly checkable witness. A solver's bare UNSAT response is not a certificate; a negative claim must freeze the instance, retain an LRAT, DRAT, VeriPB, or equivalent proof, and check that proof independently. Global S_5 symmetry lets any successful factor be renamed to 01, so fixing Y_{01} loses no existential generality.

9.3 Typed cap interfaces across a three-edge cut

A *rooted cap* (A, z, r) is a loopless cubic graph with a distinguished cap vertex z, whose three incident physical ports are a, b, c, and a proper root edge r not incident with z. Suppose a factor circuit Y_P, for $P \in \binom{[5]}{2}$, contains r and exactly ports a, b. Port c is inactive. Since $|P| = |q(c)| = 2$, exactly one of

$$P = q(c), \qquad P \cap q(c) = \varnothing \tag{16}$$

holds; call these modes in and out, respectively. The *typed signature* $\mathcal{T}(A, z, r)$ is the subset of

$$\{ab, ac, bc\} \times \{\mathsf{in}, \mathsf{out}\} \tag{17}$$

realized by some D_5-flow, factor, and root circuit.

After normalizing the ordered connector triangle to

$$q(a) = 01, \qquad q(b) = 02, \qquad q(c) = 12, \tag{18}$$

the possible factors using ports a, b are

$$12, \quad 03, \quad 04. \tag{19}$$

The first is internal, and the last two are external and exchanged by the unused-coordinate transposition $3 \leftrightarrow 4$, which fixes (18).

Proposition 31 (typed-port gluing)**.** *Cut a nontrivial three-edge cut of a cubic graph and cap each shore by a new cubic vertex, preserving the identification of the three physical ports. Let r, s be proper roots on the two shores. If the two rooted cap signatures share a typed state in* (17)*, then the glued graph has a D_5-flow in which one factor circuit contains both roots.*

Proof. Choose witnesses for the common physical port pair and common mode. Permute coordinates independently on the two shores so their ordered connector triangles both equal (18). In internal mode (19) makes the factor the unique inactive connector label, so the factor pairs agree. In external mode the factor is 03 or 04; if the choices differ, apply $3 \leftrightarrow 4$ on the second shore. This fixes all three connector labels and makes the factor pairs equal.

Delete the cap vertices and glue corresponding semiedges. Equal connector labels give a D_5-flow across the cut. On each shore, deleting the cap vertex opens the displayed factor circuit into a factor path through its root with endpoints at the same two physical ports. Gluing the two paths produces one factor circuit through r, s. ☐

Call a signature a *double star* if, for one physical port, both modes occur on each of the two port pairs incident with it. Two double stars on a three-port set share a physical pair and both modes there. Therefore:

Corollary 32 (double-star gluing)**.** *Two rooted caps whose exact typed signatures each contain a double star always glue root-good. More generally, the same holds if one signature contains a double star and the other signature covers all three physical ports.*

Proof. For the first statement, two two-edge stars in the three-edge port triangle intersect, and both signatures contain both modes on a common pair. For the second, any set of port pairs covering all three ports has at least two members, so it meets the two pairs of the double star; on that common pair the double-star shore offers whichever mode the other shore realizes. Apply Proposition 31. ☐

9.4 Exact cut relations and complement translations

The six-bit signature above deliberately unions states realized by different flows. For composition it is useful to retain the correlations within one flow. Let a two- or three-edge cut separate a proper root edge r from the cap vertex z, and suppose the cut is disjoint from the three cap edges. Cut the edges into corresponding ordered semiedges on a root pole R and a cap pole Z. Xoring all vertex equations on either shore shows that the boundary labels xor to zero. Up to a coordinate permutation, the ordered boundary word is therefore

$$(01, 01) \quad \text{for a two-cut,} \qquad (01, 02, 12) \quad \text{for a three-cut.} \tag{20a}$$

For a fixed pair P, either zero or two boundary semiedges are active in Y_P. In the latter case the active subgraph on either pole has a unique path between those semiedges.

For a partial flow q_R, let $\rho(q_R)$ be the set of factors P whose boundary path contains r. For a partial cap flow q_Z and a physical cap pair $j \in \{ab, ac, bc\}$, let $\epsilon_j(q_Z)$ be the set of factors whose boundary path contains z, uses physical pair j, and is external at z. Retain the exact relations

$$\mathcal{R} = \{\rho(q_R) : q_R \text{ has the fixed boundary word}\}, \qquad \mathcal{E} = \{(\epsilon_{ab}, \epsilon_{ac}, \epsilon_{bc})(q_Z) : q_Z \text{ has that word}\}. \tag{20b}$$

Theorem 33 (rooted cut natural join)**.** *Partial flows on the two poles with the same ordered boundary word glue bijectively to whole D_5-flows having that word on the cut. For a glued pair (q_R, q_Z), the external physical-pair mask is exactly*

$$\big\{j : \rho(q_R) \cap \epsilon_j(q_Z) \neq \varnothing\big\}. \tag{20c}$$

Consequently, one glued flow externally covers all three physical ports if and only if some $(\rho, \epsilon) \in \mathcal{R} \times \mathcal{E}$ makes at least two of the three intersections in (20c) *nonempty.*

Proof. Matching boundary labels turn each corresponding semiedge pair back into one labelled edge. All vertex equations were already satisfied, so this is a whole flow; restriction is the inverse operation.

Fix P. If the whole Y_P-circuit through r crosses the cut, its restriction to R is the unique boundary path, and hence $P \in \rho(q_R)$. The circuit reaches z in external physical mode j precisely when its restriction to Z is the boundary path recorded by $P \in \epsilon_j(q_Z)$. Conversely, two such paths have matching active boundary semiedges and glue to one factor circuit through r, z. This proves (20c). Two distinct pairs among ab, ac, bc cover all three ports, proving the last assertion. ☐

For existence questions, inclusion-dominated relation rows may be deleted: if $\rho \subseteq \rho'$, then ρ' succeeds against every cap row against which ρ succeeds, and the analogous assertion holds coordinatewise for cap triples. Keeping only inclusion-maximal rows is therefore an exact antichain compression. It is not sound to replace a relation by the coordinatewise union of rows realized by unrelated flows.

There is also a legal translation not generated by ordinary pairwise Kempe switches. It is elementary, but it changes the relevant flow orbit.

Lemma 34 (complement translation)**.** *Let q be a D_5-flow, fix $h \in [5]$, and let C be an Eulerian edge-subset all of whose labels avoid h. Put $S = [5] \setminus \{h\}$ and define*

$$q'(e) = \begin{cases} q(e) \triangle S, & e \in C, \\ q(e), & e \notin C. \end{cases} \tag{20d}$$

Then q' is a D_5-flow. For every $P \in D_5$,

$$Y_P(q') = Y_P(q) \triangle C \iff h \in P, \tag{20e}$$

while $Y_P(q') = Y_P(q)$ when $h \notin P$.

Proof. Every old label on C is a two-subset of the four-set S, so its symmetric difference with S is again a two-subset. At each vertex an even number of incident labels is changed by the same vector S, so the flow equation is preserved. For $e \in C$, membership in Y_P changes by $|S \cap P| \bmod 2$. Since $|P| = 2$, this parity is one exactly when $h \in P$, proving (20e). ☐

Proposition 35 (constant circuit translations)**.** *Suppose a fixed vector $t \in \mathbb{F}_2^5$ is added to every label on a circuit, and every old and new label has weight two. Apart from $t = 0$, exactly two cases are possible:*

1. *$|t| = 2$ and every circuit label crosses t; the circuit is a component of Y_t, so this is an ordinary component Kempe switch;*

2. *$|t| = 4$ and every circuit label avoids the coordinate outside t; this is Lemma 34.*

Proof. For every old label A,

$$|A \triangle t| = 2 + |t| - 2|A \cap t| = 2.$$

Thus $|t|$ is even; in five coordinates only $0, 2, 4$ remain. If $|t| = 2$, the equation forces $|A \cap t| = 1$, so every circuit edge is active in Y_t. At every circuit vertex those two edges exhaust the active degree, and the circuit is a component. If $|t| = 4$, the equation forces $|A \cap t| = 2$, so $A \subseteq t$, which is exactly the complement case. □

The affine description behind the lemma is sometimes useful. With $H_h = \{e : h \notin q(e)\}$, write

$$Y_{hi} = A \cup B, \quad A = \{e : h \in q(e),\ i \notin q(e)\}, \quad B = \{e : h \notin q(e),\ i \in q(e)\}.$$

All fixed-h complement translations replace B by $B \triangle C$ for an Eulerian $C \subseteq H_h$. Thus the reachable hi-parts in H_h form exactly the affine cycle-space coset of B, or equivalently all $X \subseteq H_h$ with $\partial X = \partial B$. Turning this linear description into the required rooted connectivity theorem remains open.

A complete order-12 and order-14 audit supplies a sharp delimiter. Among 8,341,704 ordinary-Kempe-orbit/rooted-interface pairs at order 14, 72 orbits have no simultaneous external flow. A single simple complement circuit immediately repairs 60. On the remaining 12, every arbitrary Eulerian support in every fixed-h cycle space still fails immediately. All 12 nevertheless admit two successive complement translations, and each has a root/cap-disjoint five-circuit leading to an ordinary Kempe orbit that contains a simultaneous flow. Thus universal immediate one-translation rescue is false, while the exact quotient-distance-one statement survives this finite census. The two graphs supporting the 12 delimiters are Tait-colourable and have girth at most four, so they lie outside the non-Tait marked-girth cap domain.

The companion checker exhausts the literal 324 normalized flows of a two-cut example and 27 flows of a three-cut example and verifies (20c) against direct whole-graph traversal. It also checks complement translations that cross the ordinary Kempe-orbit delimiters at orders 12 and 14. A bounded pole census through order ten finds 7 root and 9 cap relations for two-cuts, and 108 root and 222 cap relations for three-cuts; all $63 + 23{,}976$ cross-pairs admit a joined flow externally covering all three ports. These are finite results.

Parity, local cap-triangle legality, and residual boundary symmetry do not by themselves imply compatibility. The complete abstract orbit audit finds a two-cut forbidden pair whose root relation is the orbit of $\{02\}$ and whose cap relation is the orbit of $(\varnothing, \{02\}, \{12\})$; all 36 joins cover at most one physical pair. For a three-cut, the singleton relations

$$\mathcal{R} = \{\{01\}\}, \qquad \mathcal{E} = \{(\varnothing, \{01\}, \{02\})\} \tag{20f}$$

join only on physical pair ac. Both patterns satisfy the stated local axioms. Neither pair occurs as the complete relations of the connected shores in the order-ten census: whenever one forbidden state orbit occurs, additional companion states repair the join. Thus the exact missing lemma must use graph realizability or switch closure to force such companions; finite Boolean algebra and S_5-invariance alone are insufficient.

Proposition 36 (typed-cap finite frontier through order 14)**.** *For every biconnected simple cubic graph generated by* `geng -Cq -d3 -D3 n`*, for every even* $n \in \{4, 6, 8, 10, 12, 14\}$*, every exact rooted cap signature contains a double star. The complete census is*

n	*graphs*	*rooted interfaces*	*failures*
4	1	12	0
6	2	72	0
8	5	360	0
10	18	2,160	0
12	81	14,580	0
14	480	120,960	0
total	587	138,144	0

(20)

The six mask bits, in order ab-in, ab-out, ac-in, ac-out, bc-in, and bc-out, take only the values $15, 47, 51, 59, 60, 62, 63$ *once order ten is reached. The inclusion-minimal masks* $15, 51, 60$ *are exactly the three four-state double stars.*

This proposition is a finite computer-assisted statement. The C++ producer enumerates every D_5-flow modulo global S_5 and every rooted interface through order 14. A separately structured Python program recomputes the full census only through order 10 and independently exhausts the six-state gluing algebra; orders 12 and 14 do not yet have a second full implementation. Most importantly, the universal double-star premise is *open*. Proposition 36 therefore proves neither universal three-cut reducibility, the rooted-feasibility premise, nor FiveCDC.

Proposition 37 (one-flow external frontier through order 14)**.** *For every rooted interface in Proposition 36, there is one individual* D_5*-flow whose external root-factor components collectively meet all three physical cap ports. Equivalently, that one flow realizes external states on at least two of the three physical port pairs. The exact all-flow counts are*

n	*graphs*	*flows/*S_5	*interfaces*	*failures*
4	1	2	12	0
6	2	13	72	0
8	5	128	360	0
10	18	1,525	2,160	0
12	81	25,960	14,580	0
14	480	537,418	120,960	0

This strengthens the aggregate statement at the same finite orders: all three ports are covered in one flow, rather than by states collected from unrelated flows. It is sufficient for typed gluing. Indeed, each shore then offers at least two of the three physical port pairs, so the two sets intersect; both states are external, and a permutation of the two unused coordinate names aligns their factor pairs without disturbing the ordered cap triangle.

The quantifier is over all flows, not over every fixed flow or every Kempe orbit. Explicit small orbits fail the corresponding orbitwise claim. A C++ census proves the displayed frontier, a structurally separate Python enumeration independently reaches order 12, and a third orbit-based audit independently rechecked order 14. No finite cutoff principle is known, so the universal one-flow premise remains open and this proposition does not resolve FiveCDC.

Proposition 38 (certificate-checked extended interface frontier)**.** *Every proper cap/root interface satisfies the double-star, aggregate-external, and one-flow simultaneous-external predicates on each of the following finite inputs:*

input	*graphs*	*interfaces*	*flows*
all biconnected cubic, order 16	3,874	1,301,664	65,185
cyclic-4 non-Tait, orders 18–26	1,491	1,361,316	57,899
seven retained strong-snark rows, order 34	7	11,424	997
total	5,372	2,674,404	124,081

(21)

The first row is regenerated by `geng` with the documented order-16 options. The second uses the retained Snarkhunter canonical streams at orders 18, 20, 22, 24, and 26; their completeness depends on the documented generator boundary. The order-34 row is deliberately a claim about seven literal retained graphs, not a complete strong-snark census. The primary search emits only the explicit flows needed by the predicates. A separately written Python checker decodes every graph, verifies every weight-two label and vertex xor, reconstructs all factor components, checks the graph premises, and exhausts all interfaces from those witnesses. This is a finite positive certificate, not a cutoff or a universal theorem.

9.5 The relevant-cap cutoff

The unrestricted cap census above does not yet enter the short-cycle geometry of a hypothetical minimum obstruction. That geometry gives a substantial analytic cutoff.

Theorem 39 (relevant-cap cutoff)**.** *Let G be simple, cubic, cyclically 4-edge-connected, and of girth at least ten. Eliminate an edge to obtain $H = G \div e$, with independent artificial roots r, s. If H has a nontrivial three-edge cut, then:*

1. *neither root crosses the cut, and the roots lie on opposite shores;*
2. *each capped shore A is simple and 3-edge-connected;*
3. *for the cap vertex z, the core $K = A - z$ is connected and bridgeless, has degree sequence $2^3 3^{n-3}$, satisfies $g(K) \geq 9$ and $g(K - r) \geq 10$, and has $n \geq 57$; and*
4. *each cap has order at least 58, and $|V(G)| \geq 116$.*

Proof. The elimination lemma makes H simple and 3-edge-connected. Each shore X of a nontrivial three-cut contains a circuit, because

$$|E(H[X])| = (3|X| - 3)/2 > |X| - 1.$$

Let t roots cross the cut. Restoring the two deleted vertices replaces each crossing root by exactly one crossing spoke, regardless of which shore receives its inserted vertex. If $t = 2$, put both inserted vertices on one shore; one old cut edge and two spokes give a cyclic three-cut of G. If $t = 1$, put both vertices with the noncrossing root; two old cut edges and one spoke give such a cut. If $t = 0$ and both roots lie on one shore, put both vertices there and retain the original three-cut. The shore circuits survive these replacements, contradicting cyclic 4-edge-connectivity. Hence the roots are noncrossing and separated.

No two cut edges share a shore endpoint: deleting such an endpoint from its nontrivial shore would leave a nonempty set with edge boundary at most two in H. Thus capping creates no parallel edge. Any cut of size at most two in a cap gives the same cut in H, after taking the shore not containing the cap vertex; hence the cap is 3-edge-connected. Deleting the cap vertex leaves one degree-two vertex at each port and degree three elsewhere. It is connected, since otherwise one component uses at most one port. If an edge of the core were a bridge and one bridge shore contained k ports, the two shores would have cap boundaries $1 + k$ and $1 + (3 - k)$; both cannot be at least three. Thus the core is bridgeless, and in particular $K - r$ is connected.

A circuit of K avoiding the shore root lifts to G without changing length; one using the root gains one edge. Therefore $g(K) \geq 9$, and every 9-circuit uses r, equivalently $g(K - r) \geq 10$.

It remains to bound $n = |V(K)|$. Put $L = K - r$ and $m = |E(L)| = (3n - 5)/2$. Every degree in L lies in $\{1, 2, 3\}$, and, with $\delta(v) = 3 - d_L(v)$,

$$\sum_v \delta(v) = 5, \qquad \sum_v \delta(v) d_L(v) \leq 10. \tag{21}$$

The second inequality includes the possibility that deleting r creates a degree-one vertex. Indeed, for $d = 1, 2$ one has $(3 - d)d = 2$, and there are at most five deficient vertices.

For $j \geq 1$, let W_j be the number of oriented nonbacktracking walks of length j in L, and let $N_j(v)$ count those ending at v. By reversal and the maximum-degree-three bound,

$$N_j(v) \leq d_L(v) 2^{j-1}.$$

Extending a walk ending at v gives exactly $d_L(v) - 1 = 2 - \delta(v)$ choices. Hence

$$W_{j+1} = 2W_j - \sum_v \delta(v) N_j(v) \geq 2W_j - 10\, 2^{j-1}. \tag{21a}$$

Moreover

$$W_2 = \sum_v d_L(v)(d_L(v) - 1) = 6n - 25 + \sum_v \delta(v)^2 \geq 6n - 20.$$

Three applications of (21a) give

$$W_3 \geq 12n - 60, \qquad W_4 \geq 24n - 160, \qquad W_5 \geq 48n - 400. \tag{21b}$$

Now root a nonbacktracking ball at an edge uv of L: include u, v, and on either side include every nonbacktracking path of lengths one through four whose first edge is not uv. All displayed endpoints are distinct. A collision on one side would create a circuit of length at most eight; a collision across the two sides, together with uv, would create one of length at most nine. Both contradict $g(L) \geq 10$. Summing the resulting bound over the m choices of the root edge counts every oriented nonbacktracking walk of lengths two through five exactly once, according to its first edge and orientation. Therefore

$$mn \geq 2m + W_2 + W_3 + W_4 + W_5.$$

Substitution of $m = (3n - 5)/2$ and (21b) yields

$$3n^2 - 191n + 1290 \geq 0. \tag{21c}$$

The left side is -46 at $n = 8$, -140 at $n = 55$, and 2 at $n = 56$; its vertex lies between 31 and 32. Since K contains a circuit of length at least nine, $n \geq 9$, so (21c) forces $n \geq 56$. Finally $3n - 3$ is even, hence n is odd and $n \geq 57$. Capping adds one vertex, and restoring the two eliminated vertices gives $|V(G)| \geq 57 + 57 + 2 = 116$.

This is an edge-rooted degree-deficit refinement in the spirit of the irregular Moore bound of Alon, Hoory, and Linial [2]; the displayed count is self-contained. $\square$

Remark 40 (external extremal-data comparison). Afzaly and McKay's online extremal-graph table reports the exact maximum numbers of edges in girth-at-least-ten graphs on $57, 59, 61$ vertices as $80, 84, 87$, respectively [1]. The graph L would require $83, 86, 89$ edges at those three odd orders. Importing those external exhaustive values would therefore strengthen the conclusion to $n \geq 63$, cap order at least 64, and parent order at least 128. The source describes this as joint work with papers in preparation; this project does not retain an independently checkable exhaustion certificate. We consequently record the stronger implication as an external comparison, not as part of the self-contained theorem above.

Proposition 41 (Tait caps glue)**.** *If both rooted caps in Theorem 39 are Tait-colourable, then their typed signatures have a common external state, and the original rooted graph is root-good.*

Proof. In either cap, fix a Tait colouring. For each physical port a, 2-connectivity gives a circuit through the root and a, using one other port b. Apply the formula in Theorem 16; its factor is that circuit, while the inactive port label is disjoint from the factor pair. Thus the external states of each cap cover all three ports and consequently use at least two of the three physical port pairs. Two such pair sets intersect. On an intersecting pair both modes are external, so Proposition 31 applies. □

Proposition 42 (adjacent ports are external)**.** *Let (A, z, r) be a rooted cap carrying a D_5-flow. If the proper root edge r shares an endpoint with a physical port a, then some external factor circuit contains r and a.*

Proof. Normalize the connector triangle as

$$q(a) = 01, \qquad q(b) = 02, \qquad q(c) = 12.$$

At the common endpoint of r and a, the local-triangle lemma makes $q(r)$ one of

$$02, 03, 04, 12, 13, 14.$$

The external factor pairs through physical port a are $03, 04, 13, 14$. In the displayed root-label order, choose respectively

$$03, 04, 03, 13, 03, 04.$$

The chosen pair meets both $q(a)$ and $q(r)$ oddly, so the adjacent edges a, r lie in one factor component. At z, factors $03, 04$ use ports a, b and leave 12 inactive, while factors $13, 14$ use ports a, c and leave 02 inactive. The inactive label is disjoint from the factor pair in either case, which is exactly external mode. □

The exact surviving three-cut branch therefore has at least one non-Tait cap, and that cap has order at least 58. Minimality supplies a D_5-flow on each cap, so Proposition 42 settles any selected port adjacent to its root. Ports vertex-disjoint from the root remain open. This is a restriction, not a proof that the cap's typed signature contains a double star or even that its external states cover all three ports.

As a positive calibration, replace every Petersen vertex by the three-pole obtained from the Foster graph by deleting one vertex. The resulting Petersen–Foster graph is non-Tait, has 890 vertices and girth ten. For one cap vertex and one proper root, six literal 1,335-edge D_5-flows realize all six typed states, so the exact signature is mask 63. The companion checker verifies the graph, core restrictions, every vertex xor, and every factor component. This one-interface witness neither proves a census nor the universal double-star premise.

The full signature is an aggregate over different flows. In one retained flow the fixed typed signature has mask 6: one physical pair is external, one is internal, and the third physical port has no external state. A second checker verifies both foreign-coordinate blockers on both arcs of the internal root circuit. The cap is non-Tait and satisfies the exact 3-connectivity and marked-girth geometry above. Consequently, those blockers alone cannot force externality in an arbitrary fixed flow; an existential proof must change the flow or use additional global structure. The aggregate mask 63 means this is not a counterexample to external coverage, rooted feasibility, or FiveCDC.

There is an exact pointwise description of how a component switch repairs such a failure. If D is one component of $Y_T(q)$, and q' is obtained by transposing the two coordinates of T on D, then

$$Y_P(q') = \begin{cases} Y_P(q) \triangle D, & |P \cap T| = 1, \\ Y_P(q), & |P \cap T| \in \{0, 2\}. \end{cases} \tag{21d}$$

Indeed, outside D no label changes; on D, the transposition fixes P in the second case and replaces it by $P \triangle T$ in the first, toggling factor membership. Thus a Kempe move is literally a circuit splice, although connectivity inside the symmetric difference is still global.

The retained Petersen–Foster flow has 124 component-switch choices. Exactly six repair its missing selected port, and four of those resulting flows simultaneously cover all three ports. Two explicit eight-vertex controls show that neither selected-port externalization nor changing a fixed physical pair from internal to external is guaranteed in one switch: both first succeed after two displayed switches. These certificates motivate unrestricted flow reconfiguration while ruling out a one-switch proof; they do not establish a universal escape theorem.

9.6 Exact finite plateau results

The state enumeration quotients only by the global S_5 action. Graphs were generated canonically by `geng -Cq -d3 -D3 n`. The complete order-12 terminal-plateau audit gives:

biconnected simple cubic graphs	81
normalized D_5-flows	25,960
Kempe orbits	2,650
terminal Φ-plateaus	2,659
non-root-universal terminal plateaus	0

(22)

The terminal values range from -3 to 2, and plateau sizes range from 1 to 432. This is a complete finite census, not a proof of the terminal-plateau conjecture.

An additional support diagnostic found 420 terminal plateaus in which every state uses all five coordinates. Thus the order-12 positive result cannot be explained by always reaching a plateau with an unused coordinate and reducing to a four-coordinate argument. This support diagnostic is present in the producer report but is not among the fields recomputed by the focused independent verifier described below.

At order 14, the complete terminal-plateau computation gives:

biconnected simple cubic graphs	480
normalized D_5-flows	537,418
terminal Φ-plateaus	33,598
non-root-universal terminal plateaus	0

(23)

This includes terminal plateaus below the global maximum of an orbit.

There is a sharper finite descent test. For fixed roots r, s, let $d_q(r, s)$ be the minimum number of factor components in an intersection chain from r to s. Thus $d = 1$ is exactly root-goodness. Inside each terminal plateau, split the states of fixed distance $d > 1$ into components under neutral, distance-preserving moves. Every such component has a neutral exit to smaller distance:

terminal fixed-$d > 1$ subplateaus	642,167
maximum distance	3
subplateaus without a neutral descent exit	0

(24)

This suggests the following exact human target: every fixed-$d > 1$ subplateau in every terminal Φ-plateau has a neutral edge to smaller d. Repeated descent would prove terminal root-universality. The assertion is unproved outside the finite census.

As a separate unrestricted control, all 3,874 biconnected simple cubic graphs of order 16 were checked. Across 15,187,695 normalized flows, all 223,926,065 initially root-bad state/root pairs were rescued using only switches on factor components containing exactly one root; the maximum minimum distance was five. No graph in that order-16 census has girth eight, so this computation does not directly reach the high-girth subclass in Remark 29. All of these statements are finite evidence only.

The first complete high-girth edge-elimination control uses a cubic graph on 130 vertices that has girth ten and is cyclically 4-edge-connected. For each of its 195 edges, the reduced 128-vertex rooted instance was solved independently. All 195 are root-good; their reduced girths are nine or ten. The complete positive labels are retained, and a separate checker reconstructs every reduction, enumerates the parent's 3-edge-cuts, verifies all D_5 equations, and checks factor connectivity. This is one graph, not a universal theorem.

For the 890-vertex Petersen–Foster graph above, a second literal package checks all 1,335 edge eliminations. Every reduced 888-vertex graph has a displayed D_5-flow whose Y_{01}-component contains the two artificial roots. A solver-independent checker reconstructs every elimination, verifies reduced bridgelessness and girth nine, checks every label xor and the displayed transition path. The ordinary two-root insertion lemma then lifts each witness. This is a complete positive certificate for one named graph, not an extension to its substitution family or to arbitrary graphs.

9.7 Reproducibility and independent checks

The complete reports, producers, and focused independent verifiers are in the companion repository. From its root:

```
python3 scratch/audit_d5_surface_euler_switch.py
(cd scratch/eight-to-five-triangle-state-rigidity-20260731 && ./run_all.sh)
python3 scratch/check_d5_contraction_circuit_orbit_counterexample.py
python3 scratch/check_d4_cubic_trap_root_universality.py
sh scratch/d5-tait-prescribed-circuit-lift-20260731/run_all.sh
sh scratch/d5-tait-prescribed-circuit-lift-blind-audit-20260731/run_all.sh
python3 scratch/check_d5_capped_ladder_delay_family.py
(cd scratch/d5-root-transition-sat-20260731 && ./run_all.sh)
(cd scratch/d5-typed-cap-double-star-frontier-20260731 && ./run_all.sh)
(cd scratch/d5-simultaneous-external-frontier-20260731 && ./run_all.sh)
(cd scratch/d5-typed-cap-orbit-external-frontier-20260731 && ./run_all.sh)
(cd scratch/d5-cut-signature-natural-join-20260801 &&
 ./run_all.sh && ./run_abstract.sh && ./run_census.sh)
```

```
(cd scratch/d5-complement-circuit-external-frontier-20260801 && ./run_all.sh)
(cd scratch/d5-rooted-interface-frontier-through34-20260801 && ./run_all.sh)
(cd scratch/d5-root-pair-census-20260731 && ./run_all.sh)
(cd scratch/d5-existential-external-order58-frontier-20260731 && ./run_all.sh)
(cd scratch/d5-external-port-coverage-frontier-20260731 && ./run_all.sh)
(cd scratch/d5-external-kempe-splice-frontier-20260731 && ./run_all.sh)
(cd scratch/petersen-foster-full-typed-cap-signature-20260731 && ./run_all.sh)
(cd scratch/petersen-foster-all-eliminations-20260731 && ./run_all.sh)

python3 scratch/verify_d5_surface_chi_plateaus_order12.py
python3 scratch/verify_d5_root_euler_potential_reports.py
python3 scratch/verify_d5_root_kempe_order14_report.py
python3 scratch/verify_d5_terminal_chi_chain_lex_order14_report.py
python3 scratch/verify_d5_root_component_distance_order16_summary.py
python3 scratch/verify_d5_root_feasibility_lift13_girth10.py
```

The order-12 plateau verifier regenerates all graph identities and report arithmetic, then independently recomputes five deterministic boundary/extremal rows with a separately structured traversal. The order-14 fixed-distance verifier checks the frozen report digest, canonical graph order, per-row failure fields, and all summary arithmetic; its producer first reproduces the independent order-12 totals. The order-16 verifier checks the frozen graph identities, girth stratification, and summary totals. These report verifiers are independent semantic checks of selected structure and aggregate arithmetic, not second full enumerations; that limitation is part of the claim. The triangle-state package has two independently structured standard-library checkers. The rooted-transition package checks the semantic equivalence on deterministic instances. The typed-cap package has a complete C++ enumeration through order 14, while its independent Python enumeration stops at order 10; this asymmetry is part of Proposition 36, not omitted validation. The simultaneous-external package has a complete C++ enumeration through order 14 and a separately structured Python enumeration through order 12; a third orbit-based implementation independently rechecks its order-14 all-flow quantifier. The orbit package also retains the first failures of the stronger orbitwise double-star, simultaneous, and aggregate assertions. The cut-relation package independently enumerates partial and whole flows on one two-cut and one three-cut example, checks the natural-join mask against direct whole-graph connectivity for every word, and exhausts the bounded relation census through host order ten. Its complement-translation rows are checked directly from the displayed labels and circuits. The complement-circuit package separately performs the complete order-12 and order-14 orbit/interface census, exhausts the full fixed-coordinate cycle spaces on all 12 strict rows, checks depth-two escapes, and replays one 432-state delimiter without relying on the census traversal. The extended-interface package emits 124,081 positive flows and uses a separately written graph6/flow/component checker to exhaust 2,674,404 proper interfaces. It also verifies the stated graph properties, but canonical corpus completeness beyond regenerated order 16 remains a source-generator boundary. The order-58 package independently replays the degree-deficit profiles, the edge-rooted walk count, the exact polynomial cutoff, and scoped 18-cage controls. The splice package proves the pointwise factor identity for a component switch and replays the Petersen–Foster repairs and two depth-two delimiters. The Petersen–Foster package contains six literal flows, and its checker reconstructs the graph and validates every claimed typed component semantically.

10 What has and has not been established

The human-checkable results in this note are:

1. the D_5-flow/five-even-subgraph equivalence;
2. the state-transport and cycle-space lemmas, and the order-sharp 12-vertex obstruction to graph-dependent triangle-state compression of one supplied eight-cover;
3. the canonical properly five-coloured surface construction and Euler formula;
4. the boundary interpretation of the bichromatic factors;
5. the exact one-switch Euler-change formula;
6. the four-coordinate one-step locking lemma and the cubic D_4 trap theorem;
7. the construction from a Tait colouring that prescribes any circuit as one factor, proves root universality in the 2-connected case, and supports explicit inverse insertion;
8. the 30,858-flow certificate for independent root universality through order 28, conditional on the documented canonical non-Tait source boundary;
9. the capped-ladder lower bound proving that root-component rescue has no universal constant radius; and
10. the fixed-five minimum-counterexample reduction, bridgeless edge elimination, and the rooted-feasibility implication to FiveCDC;
11. the linear rooted-transition characterization and its exact native-XOR and CNF size counts; and
12. the typed-port gluing proposition and double-star corollary; and
13. the exact separated-cut natural join, antichain compression, and complement-translation lemma; and
14. the one-flow simultaneous-external frontier through order 14, with its all-flow/orbitwise quantifier boundary; and
15. the certificate-checked extended rooted-interface frontier on the complete order-16 census and the explicitly scoped retained streams through order 34; and
16. the relevant-cap order-58 cutoff, Tait-cap gluing theorem, adjacent-port external-state lemma, and the six-flow full-signature Petersen–Foster control.

The cube topology change is a short explicit certificate. The 12-state contraction trap, typed-cap order-14 frontier, and plateau tables are finite computer-assisted results with full instances, producers, and the independent-checking scope stated above.

No proof or disproof of the standard FiveCDC has been obtained. No orientable FiveCDC result has been obtained. The terminal-plateau conjecture and the universal typed-cap double-star and simultaneous-external premises are proof targets suggested by the computations, not theorems. A responsible public submission would require independent checking by a graph theorist, a broader literature comparison, and archival publication of the exact code and reports.

AI-use disclosure

This project used OpenAI Codex agents extensively under human direction. The agents proposed the D_5 and coloured-surface viewpoints, formulated the triangle-state and rooted-transition models, identified the typed-cap internal/external obstruction and adjacent-port table, derived candidate lemmas and proofs, derived the exact cut relation and complement translation, wrote and ran the enumeration programs, searched for counterexamples to intermediate claims, produced the finite certificates, and drafted this manuscript. In particular, agents found the order-12 triangle-state witness, proved the typed gluing lemma, and implemented both the C++ and Python cap censuses. Agents also designed the compact rooted certificate, generated its 30,858 flows, extended the interface certificate with 124,081 explicit flows, derived the relevant-cap walk bound, and found the Petersen–Foster typed witnesses. The semantic checker rejected an early exploratory Petersen–Foster witness whose labels and edges used different orderings; that witness was discarded, the labels were remapped by edge identity, and the six published flows were checked afresh. The human operator selected the research objective and directed the work. At the date of this draft, the manuscript has not received independent expert verification or peer review.

AI systems are not listed as authors. Any human who circulates or submits this note must personally verify the mathematical claims, take responsibility for the text and computations, confirm compliance with the venue's AI policy, and replace the working-draft author line with the responsible author list. The disclosure is intentionally broad: it does not characterize the AI as a mere copy editor.

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